Modular construction of sound-and-complete simplifiers for varieties of semi-rings

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Algebraic simplification

represent **equivalence classes** of expressions

$$1 + 3x + 6y + (-1) \in [0, 3, 6] \ni 3y + 3x + 3y \qquad \left(\text{integers mod } \begin{cases} \text{commutativity} \\ \text{associativity} \\ \text{neutrality} \end{cases}\right)$$

$$\mathbb{Z} \times \mathbb{N} \times \mathbb{N} \ni [a, n, k] \qquad \text{represents} \qquad a + nx + ky$$

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Robust code generation

- **Partial evaluation** (Carette et al., 2009, tagless final)
  esp. **partially-static data** structures (Yallop et al., 2018, frex)
- Rôte-proof **synthesis** (Allais et al., 2025, Agda/Idris frex)
- Program **synthesis** (Shepherd et al., 2026)

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Extensions beyond ordinary algebra

- **NbE** with algebraic structure (Corbyn et al., 2022)
- 2nd and 3rd **Futamura** projections of **System T** (Brown, 2027, thesis)
  uses frex for **second-order** algebraic theories

Modern algebra's core idea:

Theorem (extensional normalisation)

Let:

- $\mathcal{T}$ *be an algebraic theory;*
- $A = \langle \underline{A}, A[\![-]\!] \rangle$ *be a free* $\mathcal{T}$*-model with variables from* X*, i.e.: and* $\mathrm{var} : X \to \underline{A}$*.*

Then:

an equation $X \vdash t \equiv u$ *is* **provable** *in* $\mathcal{T}$	$\iff$	A **validates** *it* $A[\![t]\!]\,\mathrm{var} = A[\![u]\!]\,\mathrm{var}$

Takeaway

computable/effective representation of free models

$\Downarrow$

normalisation/algebraic simplification **procedures**

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NB: consequences beyond algebraic simplification

- **algebraic effects** in semantics (Plotkin and Power, 2002)
- **effect handlers** in functional programming (Bauer and Pretnar, 2015)
- **modular abstract syntax** with binding+substitution+holes (Fiore et al., 1999)

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NB2: Monad connection

For $\mathbf{F}_{\mathcal{T}}X = \langle MX, [\![-]\!] \rangle$ free $\mathcal{T}$-model over X via $\mathrm{var} : X \to MX$:

- MX is a **monad** with var its unit;
- $\mathbf{F}_{\mathcal{T}}X$ are all the **algebraic effects** of it;
- $\mathcal{T}$-models are all the **effect handlers** of them.

theory of algebraic simplifiers $\xrightarrow{\text{ought to give}}$ **building blocks** for algebraic simplifiers?

Initial results (Allais et al., 2023)

- Free involutive $\mathcal{T}$-models over/extensions by X from those for $\mathcal{T}$ and Bool $\times X$
- Free $\mathcal{T}$-homomorphism graph models/extensions over X from those of $\mathcal{T}$ for all Y

This talk

distributive combinations of theories and their simplifiers

Semiring $A = \langle \underline{A}, 1, (\cdot), 0, (+) \rangle$

- monoid $\langle \underline{A}, 1, (\cdot) \rangle$ that
- **distributes** over a commutative monoid $\langle \underline{A}, 0, (+) \rangle$:

$$0 \cdot x = 0 = x \cdot 0 \qquad (x + y) \cdot z = x \cdot z + y \cdot z \qquad x \cdot (y + z) = x \cdot y + x \cdot z$$

Semi-rings are what makes matrices 'work'.

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Can vary the two algebraic structures to get:

- rings, i.e., addition has inverses;
- (semi-)rings without multiplicative unit or additive zero;
- commutative (semi-)rings;
- involution over multiplication; etc.

We'll consider the general situation.

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Semi-rings are what makes matrices 'work'.

Distributive combination $(\mathcal{T} \triangleleft \mathcal{S})$-model $A = \langle \underline{A}, \mathcal{T}[\![-]\!], \mathcal{S}[\![-]\!] \rangle$

- $\mathcal{T}$-model $\langle \underline{A}, \mathcal{T}[\![-]\!] \rangle$; that
- **distributes** over an $\mathcal{S}$-model $\langle \underline{A}, \mathcal{S}[\![-]\!] \rangle$, i.e., for all:
  $\mathcal{T}$-operation $(f : n) \in \mathcal{T}$, $\mathcal{S}$-operation $(g : k)$ and position $1 \leq i \leq n$:

$$\begin{aligned} &\mathcal{T}[\![f]\!](x_1, \ldots, x_{i-1}, \mathcal{S}[\![g]\!](y_1, \ldots, y_k), x_{i+1}, \ldots, x_n) = \\ &\quad \mathcal{S}[\![g]\!](\mathcal{T}[\![f]\!](x_1, \ldots, x_{i-1}, y_1, x_{i+1}, \ldots, x_n), \ldots, \\ &\qquad \mathcal{T}[\![f]\!](x_1, \ldots, x_{i-1}, y_k, x_{i+1}, \ldots, x_n)) \end{aligned}$$

Example distributive combinations

Multiplicative structure			Additive structure		
Theory	**Comm.**[1]	**Inv.**[3]	**Comm. Semigroup**	**Comm. Monoid**	**Abelian Group**
Semigroup	**Mere**	**Mere**	$\mathcal{P}(\Sigma^+) \setminus \{\varnothing\}$	$\mathcal{P}(\Sigma^+)$	$\mathbb{Z}\langle\Sigma^+\rangle$
		Inv.[4]	$(\mathcal{P}(\Sigma^+) \setminus \{\varnothing\}, -^R)$	$(\mathcal{P}(\Sigma^+), -^R)$	$(\mathbb{Z}\langle\Sigma^+\rangle, \mathrm{rev})$
	Comm.[2]	**Mere**	$2\mathbb{N}_{>0}$	$2\mathbb{N}$	$2\mathbb{Z}$
		Inv.	$\mathfrak{m}_n^\sigma \setminus \{0\}$	$\mathfrak{m}_n^\sigma$	$(2\mathbb{Z}[\mathbf{i}], \overline{-})$
Monoid	**Mere**	**Mere**	$\mathcal{P}(\Sigma^*) \setminus \{\varnothing\}$, $M_n(\mathbb{N}_{>0})$	$\mathcal{P}(\Sigma^*)$, $M_n(\mathbb{N})$	$M_n(\mathbb{Z})$
		Inv.	$(M_n(\mathbb{N}_{>0}), -^T)$, $(\mathcal{P}(\Sigma^*) \setminus \{\varnothing\}, -^R)$	$(M_n(\mathbb{N}), -^T)$, $(\mathcal{P}(\Sigma^*), -^R)$, $\mathsf{Rel}(X)$	$(M_n(\mathbb{Z}), -^T)$
	Comm.	**Mere**	$\mathbb{N}_{>0}$	$\mathbb{N}$	$\mathbb{Z}$, $\mathbb{R}$, $\mathbb{C}$
		Inv.	$\mathbb{N}_\sigma[X_1, \ldots, X_n] \setminus \{0\}$	$\mathbb{N}_\sigma[X_1, \ldots, X_n]$	$(\mathbb{C}, \overline{-})$, $\mathbb{Z}_\sigma[X_1, \ldots, X_n]$
Group	**Mere**	**Mere**	$T_{n,\Delta_{>0}}(\mathbb{R})$	**impossible** nontrivially	**impossible** nontrivially
		Inv.	$(T_{n,\Delta_{>0}}(\mathbb{R}), -^\dagger)$	**impossible** nontrivially	**impossible** nontrivially
	Comm.	**Mere**	$\mathbb{R}_{>0}$	**impossible** nontrivially	**impossible** nontrivially
		Inv.	$(\mathbb{C}_{>0}, \overline{-})$	**impossible** nontrivially	**impossible** nontrivially

Legend: [1]Commutativity; [2]Commutative; [3]Involutivity; [4]Involutive

Theorem (Fujii et al. 2026)

For the distributive combination $\mathcal{T} \triangleleft \mathcal{S}$, if $(\mathcal{T}, \mathcal{S})$ are **compatible**:

- *Let A be free $\mathcal{T}$-model on X; and*
- *Let B be free $\mathcal{S}$-model on $\underline{A}$.*

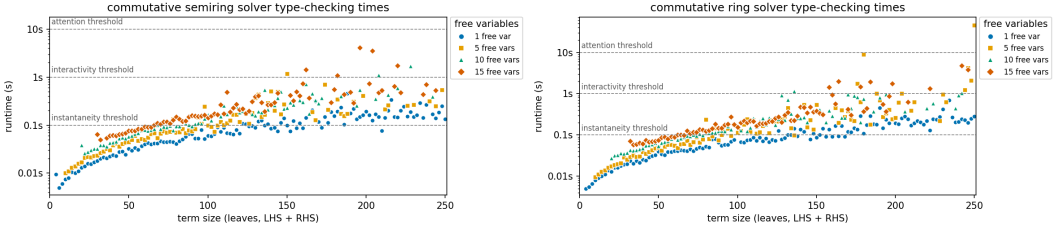
Then B equipped with **$\mathcal{S}$-convolution** *of A is the free $(\mathcal{T} \triangleleft \mathcal{S})$-model on X.*

giving highly modular algebraic simplifiers

Especially for rôte-proof synthesis, this theorem gives dozens of **sound-and-complete** algebraic simplifiers.
Modularity is an issue, c.f. Rocq's ecosystem:

- uses **almost rings** (Grégoire and Mahboubi, 2005)
- one algebraic structure
- incomplete, but good enough, simplifier

Associated implementation doesn't suck for, e.g., responsive **type-level computation**:



Type-checking times for the commutative semiring and ring simplifiers.

Talk structure

- Introduction
- Why is it difficult?
- Generalised convolution
- Verbal substructure
- Compatibility

Known instances

$$T_{\text{semi-ring}} = T_{\text{commutative monoid}} \circ T_{\text{monoid}}$$

cf. category-theoretic concepts of **monad-monad distributive law** (Beck, 1969)

Monad-connection — notable interest for PL-semanticists

Known instances

$$T_{\text{semi-ring}} = T_{\text{commutative monoid}} \circ T_{\text{monoid}}$$

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Monad-connection — notable interest for PL-semanticists

Impossibility results

Plotkin's counter-example (Varacca and Winskel, 2006)—**no** distributive law between

- probabilistic non-determinism: free Barycentric algebra / finite probability distribution monad
- non-determinism: free finite powerset monad

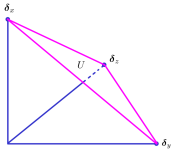
Systematically generalised by Zwart and Marsden (2022).

Distributive combination

Can be non-modular:

free (probability $\triangleleft$ non-determinism)-model carried by **credal sets** (Levi, 1980):

(finitely-generated) convex subsets of probability distributions



This approach

- Pioneered by Parlant (2020); Parlant et al. (2020); Dahlqvist et al. (2018)
- Fujii et al. (2026) connect it to Tronin's (2002) **verbal categories**, making the approach general and systematic

Limitations

- Only for monads over sets, cf. Dash and Staton's (2021) distributive combination for point processes:

  probability distributions over measurable spaces
  distributes over
  multisets / commutative monoids

- **inapplicable** theory when $\mathcal{S}$ is not **commutative**/**exchangeable**

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Synthetic convolution (folklore)

Given:

- monad $\mathcal{M} = \langle M, \mathbf{val}, \mathbf{let} \rangle$:

$$\frac{X \text{ set}}{MX \text{ set}} \qquad \frac{x : X}{\mathbf{val} x : MX} \qquad \frac{a : MX \qquad \times : X \vdash b : MY}{\mathbf{let} \times = a \mathbf{\ in\ } b : MY}$$

- operation $f : X_1 \times \cdots \times X_n \to Y$

the $\mathcal{M}$-**convolution** of f:

$$\begin{aligned} &\mathbf{Conv}_{\mathcal{M}} f : MX_1 \times \cdots \times MX_n \to MY \\ &\mathbf{Conv}_{\mathcal{M}} f(a_1, \ldots, a_n) \coloneqq \mathbf{let}\ \times_1 = a_1 \mathbf{\ in} \\ &\qquad\qquad\qquad\qquad\qquad\qquad \vdots \\ &\qquad\qquad\qquad\qquad\qquad\qquad \mathbf{let}\ \times_n = a_n \mathbf{\ in\ val} f(x_1, \ldots, x_n) \end{aligned}$$

Lift $A = \langle \underline{A}, [\![A]\!] - \rangle$ to $\mathbf{Conv}_{\mathcal{M}} A = \langle M\underline{A}, [\![A]\!]\, f \coloneqq \mathbf{Conv}_{\mathcal{M}} f \rangle$

The generalised convolution, and its converse the **de-convolution** are common:

area	operation	convolution	de-convolution
scanning/lexing/parsing	string concat	language concat	Brzozowski derivative
database theory	record concatenation	cartesian product	(some) queries
probability & statistics	trace concatenation	generative modelling	disintegration

also, collection programming with semi-ring dictionaries (Shaikhha et al., 2022).

Why 'convolution'?

Take:

- the Giry monad $DX = \{\text{probability measures over measurable space } X\}$
- $f : (+) : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$

For probability measrues $\mu, \nu \in D\mathbb{R}$ with probability density functions $\varphi, \psi : \mathbb{R} \to [0, \infty]$

(so $\mathbb{P}_\mu[E] = \int_E \mathrm{d}x\varphi\, x$ and $\mathbb{P}_\nu[E] = \int_E \mathrm{d}x\psi\, x$)

we have that $\mathbf{Conv}_{\mathsf{Giry}}(\mu, \nu)$ has density function:

$$\varphi * \psi : \mathbb{R} \to [0, \infty] \qquad (\varphi * \psi)t \coloneqq \int_{-\infty}^{+\infty} \mathrm{d}x\varphi x \cdot \psi(t - x)$$

and $\varphi * \psi$ is determined almost surely by $\mathbf{Conv}_{\mathsf{Giry}}(\mu, \nu)$, and uniquely for continuous density functions.

Central question

Given:

- algebraic structure A; and
- monad $\mathcal{S}$

which equations lift from A to $\mathbf{Conv}_{\mathcal{S}}A$?

$$A \models X \vdash l \equiv r \implies \mathbf{Conv}_{\mathcal{S}}A \models X \vdash l \equiv r$$

Example & non-example

For group $A = \langle \mathbb{Z}, 0, (+), (-) \rangle$, $\mathbf{Conv}_{\mathsf{powerset}}A$ is a monoid:

- monoid axioms lift: $0 + \mathsf{x} = \mathsf{x} = \mathsf{x} + 0 \qquad \mathsf{x} + (\mathsf{y} + \mathsf{z}) = (\mathsf{x} + \mathsf{y}) + \mathsf{z}$
- inverse axioms fail to lift: $\mathsf{x} + (-\mathsf{x}) = 0$:

$$\{1, -1\} \oplus (\ominus \{1, -1\}) = \{1, -1\} \oplus (\{-1, 1\}) = \{0, 2, -2\} \neq \{0\}$$

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$$A \models X \vdash l \equiv r \implies \mathbf{Conv}_{\mathcal{S}}A \models X \vdash l \equiv r$$

- Gautam (1957) and Manes and Mulry (2007):
  for $\mathcal{S} =$ powerset, exactly linear equations
- Parlant (2020) characterises for: commutative $\mathcal{S}$, commutative-affine $\mathcal{S}$, etc.
- Fujii et al. (2026) systematic theory for commutative $\mathcal{S}$

Fujii et al.'s (2026) approach

- Abstract over 'substructural manipulation of context': Tronin's (2002) **verbal subcategories** $\mathbb{W}$
- Theories $\mathcal{T}$ **definable** by $\mathbb{W}$-**allowed** axioms: $\mathcal{T} \underset{\mathbb{W}}{\perp}$
- $\mathbb{W}$-**commutative monads** $\mathcal{S}$ preserve $\mathbb{W}$-structure: $\underset{\mathbb{W}}{\perp} \mathcal{S}$.

The main theorem holds for $\mathcal{T} \underset{\mathbb{W}}{\perp} \mathcal{S}$ with $\mathbb{W}$ permitting variable exchange.

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Algebraic contexts $\Gamma, \Delta \in \mathbb{\Gamma}$

represent by natural numbers

variables in Γ are the numbers below Γ

Write: $\Gamma = \{x_1, \ldots, x_n\}$, $\Delta = \{y_1, \ldots, y_k\}$

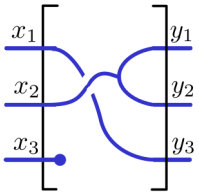
Renaming $\Gamma \vdash \rho : \Delta$

maps variables $y \in \Delta$ to variables $y\,[\rho] \in \Gamma$

Example

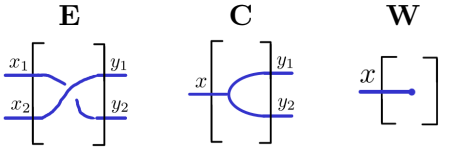
$$\{x_1, x_2, x_3\} \vdash \rho : \begin{cases} x_2 \leftarrow\!\shortmid y_1 \\ x_2 \leftarrow\!\shortmid y_2 \\ x_1 \leftarrow\!\shortmid y_3 \end{cases} : \{y_1, y_2, y_3\}$$

ρ



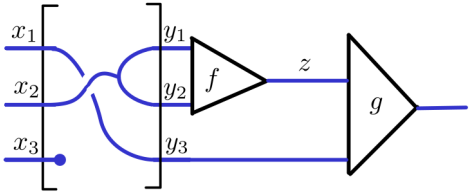
information flow graph notation

The three 'structural' rules are:

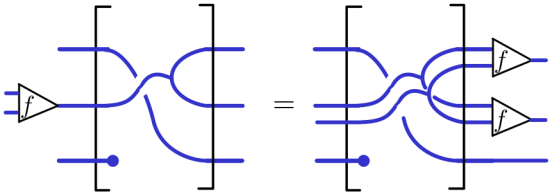


exchange contraction weakening

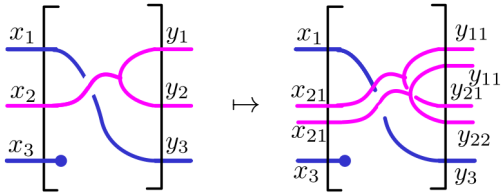
Renaming compose sequentially and 'in-parallel'. We use them to depict structural manipulation on contexts of algebraic expressions:



Verbal commutativity ('naturality')



Verbal inflation



Verbal subcategory $\mathbb{W}$ (Tronin, 2002)

$\mathbb{W} \subseteq \mathbb{\Gamma}$ closed under:

- identities
- sequential and parallel composition
- verbal inflation

Verbal closure

S_{vrb} smallest verbal subcategory containing S:

- $\{\mathbf{E}\}_{\mathrm{vrb}} = \mathbb{W}_{\mathsf{bij}}$: bijective renamings (all variable exchanges permitted)
- $\{\mathbf{E}, \mathbf{C}\}_{\mathrm{vrb}} = \mathbb{W}_{\mathsf{surj}}$: surjective renamings (all variables are relevant)
- $\{\mathbf{E}, \mathbf{W}\}_{\mathrm{vrb}} = \mathbb{W}_{\mathsf{surj}}$: injective renamings (affine use of variables)

Verbal subcategory $\mathbb{W}$ (Tronin, 2002)

$\mathbb{W} \subseteq \mathbb{\Gamma}$ closed under:

- identities
- sequential and parallel composition
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NB: András Kovács will use **thinnings**:
$\{\mathbf{W}\}_{\mathrm{vrb}}$: subset inclusions

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Fujii et al.'s (2026) account only concerns **symmetric/exchangeable** verbal subcategories—those containing $\mathbf{E}$.

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Linear terms $\Gamma \vdash\!\!\circ_\Sigma t$ over algebraic signature Σ

$$\frac{}{\{x\} \vdash\!\!\circ_\Sigma x} \qquad \frac{\Gamma_1 \vdash\!\!\circ_\Sigma t_1, \ldots, \Gamma_n \vdash\!\!\circ_\Sigma t_n}{\Gamma_1, \ldots, \Gamma_n \vdash\!\!\circ_\Sigma f(t_1, \ldots, t_n)}((f:n) \in \Sigma)$$

Verbally definable equation over $\mathbb{W}$

Equation of the form $\Gamma \vdash t\,[\rho] \equiv t'\,[\rho']$ where:

- $\Delta \vdash\!\!\circ\, t$ and $\Gamma \vdash_{\mathbb{W}} \rho : \Delta$
- $\Delta' \vdash\!\!\circ\, t'$ and $\Gamma \vdash_{\mathbb{W}} \rho' : \Delta'$

Verbally definable theory over $\mathbb{W}$ ('$\mathbb{W}$-operadic')

theory axiomatisable by verbally definable equations over $\mathbb{W}$
We then write $\mathcal{T} \underset{\mathbb{W}}{\perp}$

Theorem

Let $\mathcal{S}$ be a monad and $\mathbb{W} \ni \mathbf{E}$ a symmetric verbal subcategory. TFAE:

- *For every algebraic signature Σ and Σ-algebra A, and every $\mathbb{W}$-definable equation eq over Σ:*
$$A \models \mathrm{eq} \implies \mathbf{Conv}_{\mathcal{S}} A \models \mathrm{eq}$$
- *For every $\mathbb{W}$-renaming $\Gamma \vdash_{\mathbb{W}} \rho : \Delta$ and set X:*
$$\mathbf{Conv}_{\mathcal{S}} \rho = (\mathbf{Conv}_{\mathcal{S}} \mathrm{id})^{\rho} : (\mathcal{S}X)^{\Gamma} \to \mathcal{S}(X^{\Delta})$$

In this case we write $\underset{\mathbb{W}}{\perp} \mathcal{S}$ and call $\mathcal{S}$ a **$\mathbb{W}$-commutative** monad.

Theorem (Fujii et al. 2026; Parlant 2020)

A monad is:

- $\mathbb{W}_{bij}$*-commutative iff it is commutative.*
- $\mathbb{W}_{inj}$*-commutative iff it is commutative and affine.*
- $\mathbb{W}_{surj}$*-commutative iff it is commutative and relevant.*

(see the paper for more characterisations)

Verbally compatible theory-monad $\mathcal{T} \underset{\mathbb{W}}{\perp} \mathcal{S}$

- $\mathbb{W}$ verbal subcategory;
- $\mathcal{T}$ is $\mathbb{W}$-definable; and
- $\mathcal{S}$ is $\mathbb{W}$-commutative

Theorem (Fujii et al. 2026)

For:

- *algebraic theory* $\mathcal{T}$*;*
- *algebraic theory* $\mathcal{S}$ *(or merely monad* $\mathcal{S}$*);*
- *set* X*;*
- *free* $\mathcal{T}$*-model* $A = \langle \underline{A}, A[\![-]\!] \rangle$ *over* X *via* $\mathrm{var}^{\mathcal{T}} : X \to \underline{A}$*;*
- *free* $\mathcal{S}$*-model* $B = \langle \underline{B}, B[\![-]\!] \rangle$ *over* $\underline{A}$ *via* $\mathrm{var}^{\mathcal{S}} : \underline{A} \to \underline{B}$

If $(\mathcal{T}, \mathcal{S})$ *are* **compatible** *then have free* $(\mathcal{T} \triangleleft \mathcal{S})$*-model over* X *by:*

$$\langle \underline{B}, \mathbf{Conv}_{\mathcal{S}} A, B[\![-]\!] \rangle \qquad X \xrightarrow{\mathrm{var}^{\mathcal{T}}} \underline{A} \xrightarrow{\mathrm{var}^{\mathcal{S}}} \underline{B}$$

where $\mathbf{Conv}_{\mathcal{S}} A$ *is the* $\mathcal{S}$**-convolution** *of* A*.*

Summary

- Modular, sound-and-complete algebraic simplifiers for distributive combinations
- Suggestive performance for something free, esp. for rôte-proof synthesis
- Enabled by Fujii et al.'s (2026) analysis of verbal subcategories

Further work

- Relaxing the commutativity/exchange requirement through pre-operadic/pre-multi-categorical perspective
- Intensional normalisation via recourse to setoids
- Investigate runtime performance at scale
- What about free extensions? They require no structural axiomatisation.
- Integrate into the rest of the frex project (partial evaluation, program synthesis)

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